```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
```

```
from google.colab import files
uploaded=files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving Spotify.csv to Spotify.csv
```

```
df=pd.read_csv("Spotify.csv")
```

```
df
```

	Unnamed: 0	track_id	artists	album_name	track_name	populari
0	0	5SuOikwiRyPMVoIQDJUgSV	Gen Hoshino	Comedy	Comedy	
1	1	4qPNDBW1i3p13qLCt0Ki3A	Ben Woodward	Ghost (Acoustic)	Ghost - Acoustic	
2	2	1iJBSr7s7jYXzM8EGcbK5b	Ingrid Michaelson;ZAYN	To Begin Again	To Begin Again	
3	3	6lfxq3CG4xtTiEg7opyCyx	Kina Grannis	Crazy Rich Asians (Original Motion Picture Sou...	Can't Help Falling In Love	
4	4	5vjLSffimiIP26QG5WcN2K	Chord Overstreet	Hold On	Hold On	
...	...	...	...	...	...	
113995	113995	2C3TZjDRiAzdyViavDJ217	Rainy Lullaby	#mindfulness - Soft Rain for Mindful Meditatio...	Sleep My Little Boy	
113996	113996	1hIz5L4IB9hN3WRYPOCGPw	Rainy Lullaby	#mindfulness - Soft Rain for Mindful Meditatio...	Water Into Light	
113997	113997	6x8ZfSoqDjuNa5SVP5QjvX	Cesária Evora	Best Of	Miss Perfumado	
113998	113998	2e6sXL2bYv4bSz6VTdnfLs	Michael W. Smith	Change Your World	Friends	
113999	113999	2hETkH7cOfqmz3LqZDHZf5	Cesária Evora	Miss Perfumado	Barbincor	

114000 rows × 21 columns

```
#DATA UNDERSTANDING
```

```
print("first 8 rows")
df.head(8)
```

```
first 8 rows
```

	Unnamed: 0	track_id	artists	album_name	track_name	popularity
0	0	5SuOikwiRyPMVoIQDJUgSV	Gen Hoshino	Comedy	Comedy	73
1	1	4qPNDBW1i3p13qLCt0Ki3A	Ben Woodward	Ghost (Acoustic)	Ghost - Acoustic	55
2	2	1iJBSr7s7jYXzM8EGcbK5b	Ingrid Michaelson;ZAYN	To Begin Again	To Begin Again	57
3	3	6lfxq3CG4xtTiEg7opyCyx	Kina Grannis	Crazy Rich Asians (Original Motion Picture Sou...	Can't Help Falling In Love	71
4	4	5vjLSffimiIP26QG5WcN2K	Chord Overstreet	Hold On	Hold On	82
5	5	01MVOl9KtVTNfFiBU9I7dc	Tyrone Wells	Days I Will Remember	Days I Will Remember	58
6	6	6Vc5wAMmXdKIAM7WUoEb7N	A Great Big World;Christina Aguilera	Is There Anybody Out There?	Say Something	74
7	7	1EzrEOXmMH3G43AXT1y7pA	Jason Mraz	We Sing. We Dance. We Steal Things.	I'm Yours	80

8 rows × 21 columns

```
df.shape
```

```
(114000, 21)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 114000 entries, 0 to 113999
Data columns (total 21 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   Unnamed: 0        114000 non-null  int64
 1   track_id          114000 non-null  object
 2   artists           113999 non-null  object
 3   album_name        113999 non-null  object
 4   track_name        113999 non-null  object
 5   popularity        114000 non-null  int64
 6   duration_ms       114000 non-null  int64
 7   explicit          114000 non-null  bool
 8   danceability      114000 non-null  float64
 9   energy            114000 non-null  float64
 10  key               114000 non-null  int64
 11  loudness          114000 non-null  float64
 12  mode              114000 non-null  int64
 13  speechiness       114000 non-null  float64
 14  acousticness      114000 non-null  float64
 15  instrumentalness  114000 non-null  float64
 16  liveness          114000 non-null  float64
```

```
 17  valence           114000 non-null  float64
 18  tempo             114000 non-null  float64
 19  time_signature    114000 non-null  int64
 20  track_genre       114000 non-null  object
dtypes: bool(1), float64(9), int64(6), object(5)
memory usage: 17.5+ MB
```

```
#DATA CLEANING
#Counting the duplicate values
df.duplicated().value_counts()
```

	count
False	114000

dtype: int64

```
#Checking missing values in each column
missing=print(df.isnull().sum())
```

```
Unnamed: 0          0
track_id            0
artists             1
album_name          1
track_name          1
popularity          0
duration_ms         0
explicit            0
danceability        0
energy              0
key                 0
loudness            0
mode                0
speechiness         0
acousticness        0
instrumentalness    0
liveness            0
valence             0
tempo               0
time_signature      0
track_genre         0
dtype: int64
```

```
df["artists"]=df["artists"].fillna("unknown")
df["album_name"]=df["album_name"].fillna("unknown")
df["track_name"]=df["track_name"].fillna("unknown track")
```

```
df[["artists","album_name","track_name"]].isnull().sum()
```

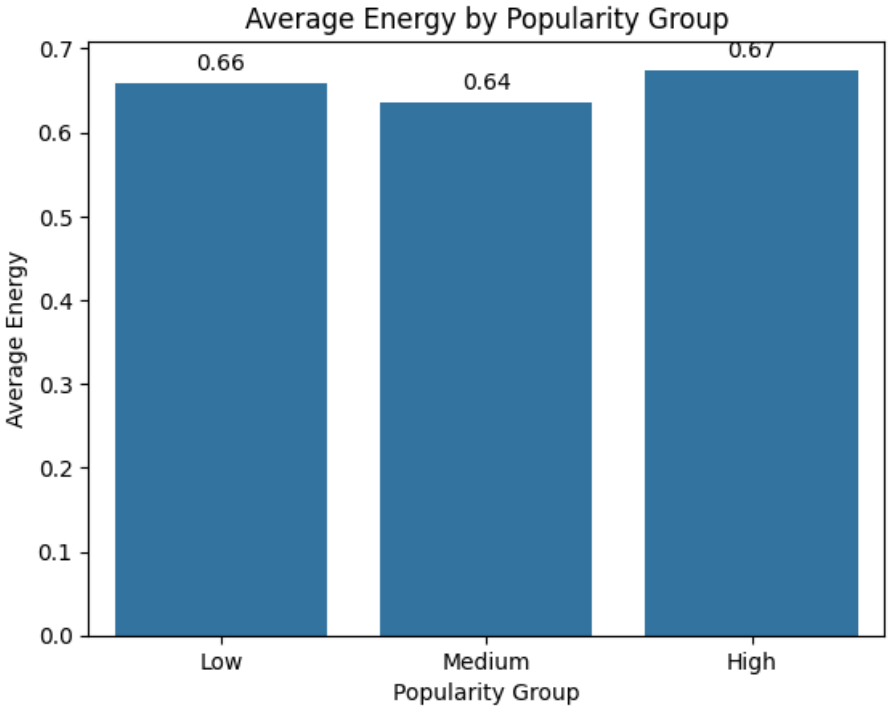
	0
artists	0
album_name	0
track_name	0

dtype: int64

```
#DATA PREPROCESSING
df_filtered = df[df['duration_ms'] < 10]
```

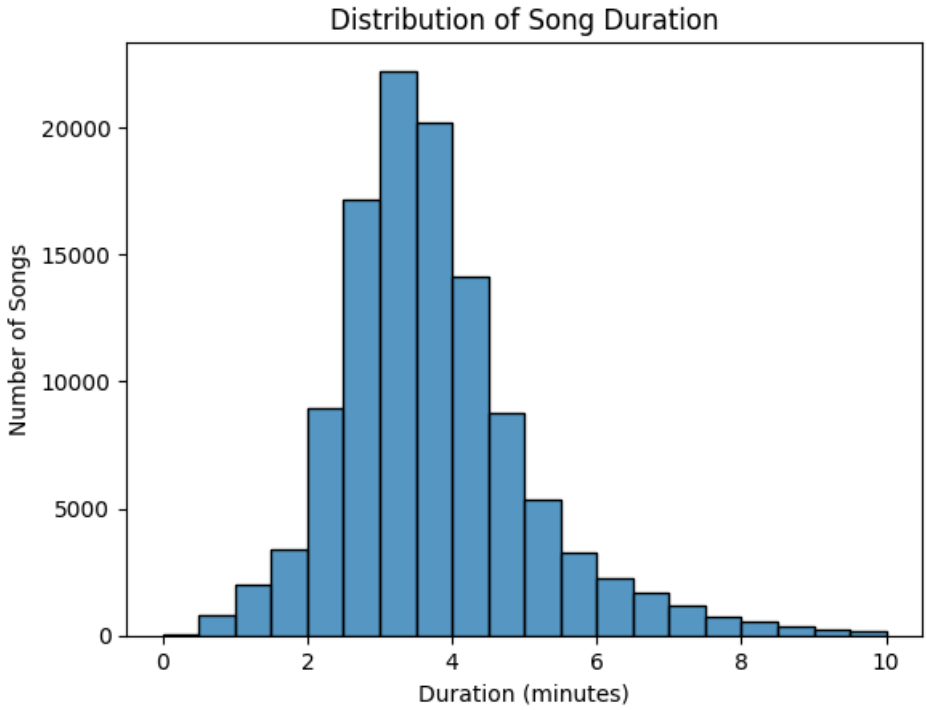
```
#DATA VISUALIZATION
#GRAPH 1 AVG ENGERY BY POPULATION
bins = [0, 30, 70, 100]
labels = ['Low', 'Medium', 'High']
df['popularity_group'] = pd.cut(df['popularity'], bins=bins, labels=labels)

grouped = df.groupby('popularity_group', as_index=False)['energy'].mean()
sns.barplot(data=grouped, x='popularity_group', y='energy')
plt.title('Average Energy by Popularity Group')
plt.xlabel('Popularity Group')
plt.ylabel('Average Energy')

for index, value in enumerate(grouped['energy']):
    plt.text(index, value + 0.01, f"{value:.2f}",
             ha='center', va='bottom')

plt.show()
print("\nObservation:")
print("""The average energy level is almost consistent across all popularity groups, wi
This suggests that song energy does not have a strong direct relationship with populari
```

```
/tmp/ipykernel_11963/1744541866.py:7: FutureWarning: The default of observed=False is de
  grouped = df.groupby('popularity_group', as_index=False)['energy'].mean()
```




```
Observation:
The average energy level is almost consistent across all popularity groups, with only mi
This suggests that song energy does not have a strong direct relationship with popularit
```

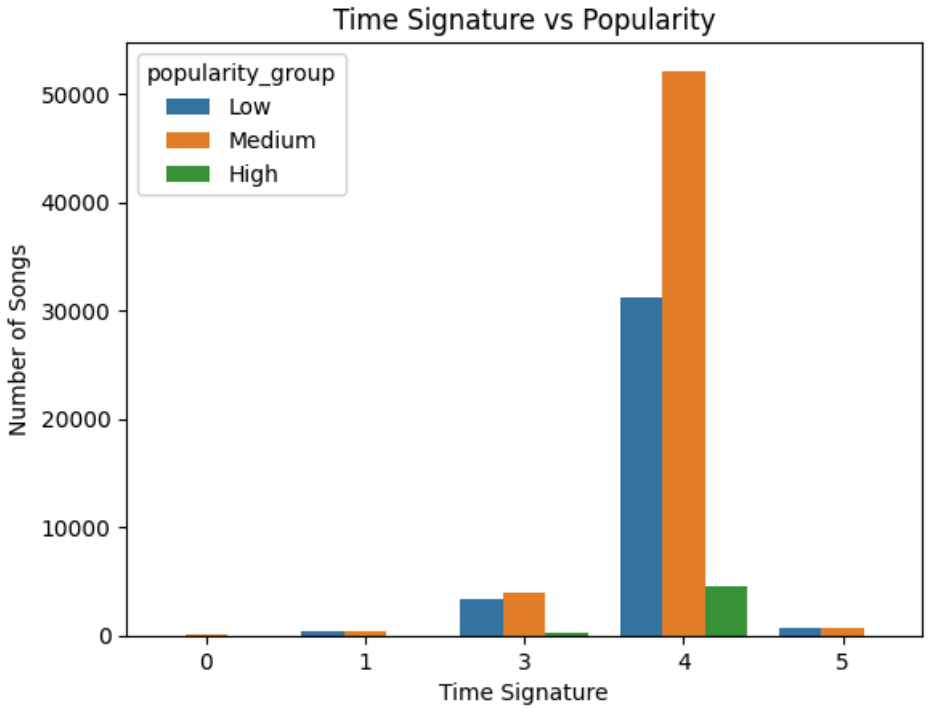
```
#GRAPH 2 SONG DURATION
sns.histplot(df_filtered['duration_ms'], bins=20, edgecolor='black')
plt.title('Distribution of Song Duration')
plt.xlabel('Duration (minutes)')
plt.ylabel('Number of Songs')
plt.show()
print("Observation:")
print("""A significant number of songs fall below 2.5 minutes
while the highest concentration of songs is observed around 3 to 4 minutes
indicating that most songs follow a standard duration range.""")
```



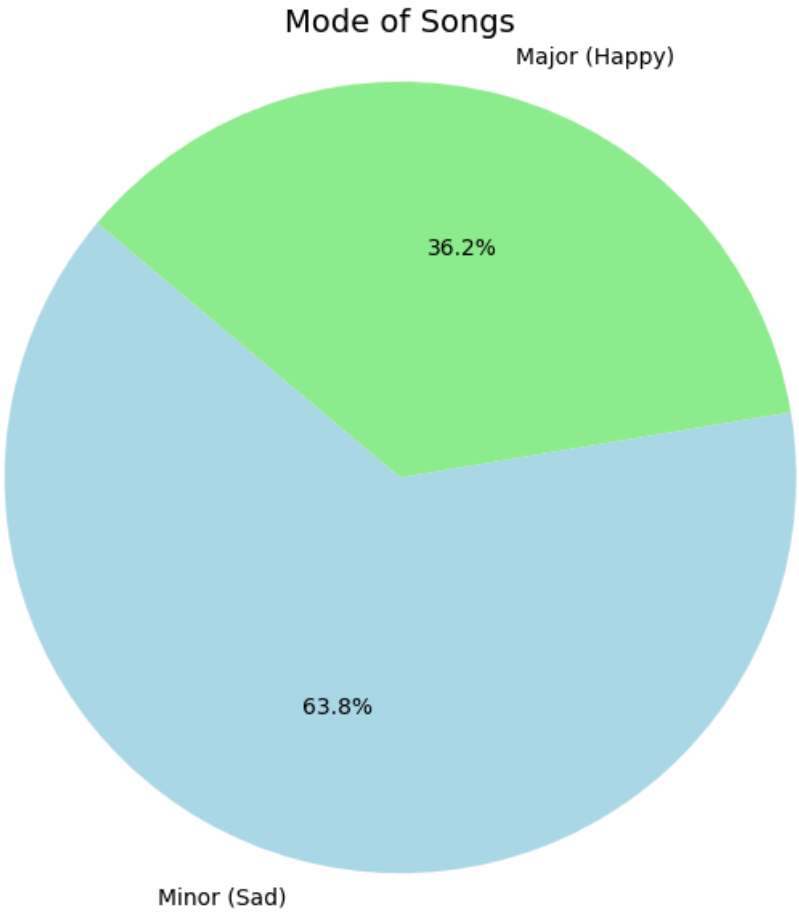

```
Observation:
A significant number of songs fall below 2.5 minutes
while the highest concentration of songs is observed around 3 to 4 minutes
indicating that most songs follow a standard duration range.
```

```
#GRAPH 3 POPULARITY OF TIME SIGNATURE
df['popularity_group'] = pd.cut(
    df['popularity'],
    bins=[0, 30, 70, 100],
    labels=['Low', 'Medium', 'High']
)

sns.countplot(x='time_signature', hue='popularity_group', data=df)
plt.title('Time Signature vs Popularity')
plt.xlabel('Time Signature')
plt.ylabel('Number of Songs')


print("Observation:")
print("Most songs have a time signature of 4 across all popularity levels")

plt.show()
```

```
Observation:
Most songs have a time signature of 4 across all popularity levels
```




```
RAPH 4 SONG MODE
de_counts = df['mode'].value_counts()

bels = ['Minor (Sad)', 'Major (Happy)']

t.figure(figsize=(7, 7))
t.pie(mode_counts, labels=labels, autopct='%1.1f%%',
      startangle=140, colors=['lightblue', 'lightgreen'])

t.title('Mode of Songs', fontsize=14)
t.axis('equal')
t.show()
int("\nObservation")
int("Most songs are in Minor mode, indicating emotional music.")
int("\nFinal conclusion :")
int("""Most songs are between 3 to 5 minutes long.
ergy levels are similar across all popularity groups,so energy alone does not decide popu
st songs use a 4-beat rhythm and are in minor mode,
owing that popularity is not strictly linked to a happy (major) sound.
erall, popularity depends on multiple factors, not just one feature.""")
```



Observation
Most songs are in Minor mode, indicating emotional music.

Final conclusion :
Most songs are between 3 to 5 minutes long.
Energy levels are similar across all popularity groups,so energy alone does not decide p
Most songs use a 4-beat rhythm and are in minor mode,
showing that popularity is not strictly linked to a happy (major) sound.
Overall, popularity depends on multiple factors, not just one feature.